DATA STORY / 01

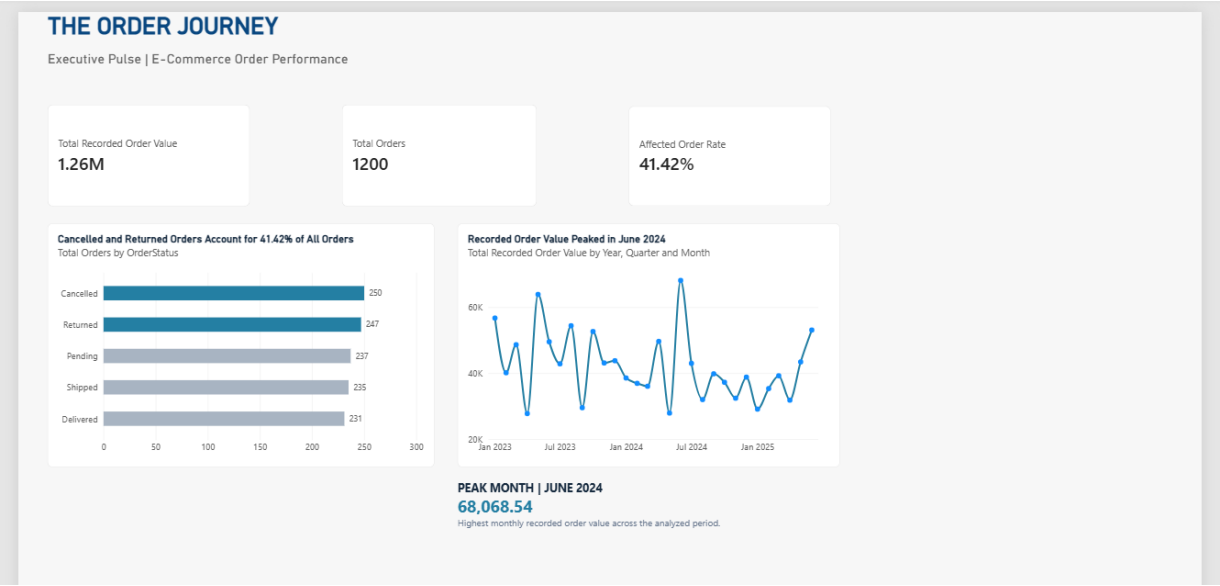
THE ORDER JOURNEY

E-Commerce Performance & Risk Intelligence

From 1,200 transactions to an evidence-led business narrative. A three-page Power BI decision experience spanning performance, customer intelligence, and operational risk.

DECODELABS • DATA ANALYTICS INTERNSHIP • PROJECT 4

POWER BI CASE STUDY / NOVEMBER 2026



Cover visual: Executive Pulse dashboard. All figures refer to the provided order dataset.

01 / EXECUTIVE BRIEF

The business question

How can an e-commerce team distinguish apparent commercial momentum from friction in the order journey? This project connects recorded value, order outcomes, customer profiles and product-payment combinations in a single analytical experience.

At a glance

METRIC	OBSERVED RESULT
Recorded order value	1,264,761.96
Recorded orders	1,200
Distinct customers	1,189
Cancelled or returned	497 orders
Affected order rate	41.42%
Coverage	January 2023 to June 2025

Three decisions, three report pages

01 Executive Pulse. Establish scale, status distribution and month-to-month order value trajectory.

02 Product & Customer Intelligence. Compare category contribution, analyst-defined customer spending segments and recorded referral sources.

03 Order Risk & Opportunities. Locate the highest observed cancellation/return concentrations and prioritize specific follow-up questions.

The pivotal observation

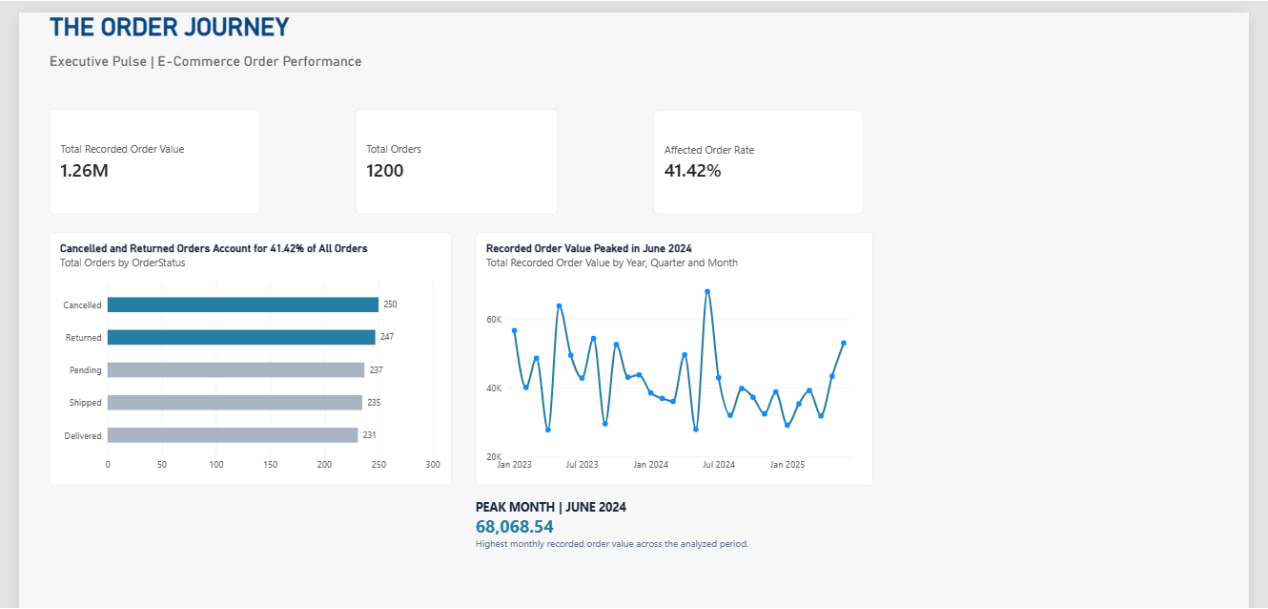
497 of 1,200 recorded orders were cancelled or returned. The resulting 41.42% rate is an operational signal, not a measured financial loss. It calls for better diagnosis of the order journey rather than an unsupported conclusion about cause.

Definitions: “Affected” = Cancelled or Returned. “Recorded order value” includes all order statuses and must not be confused with realized revenue.

02 / EXECUTIVE PULSE

Performance first

What is the overall condition of the order journey?



Dashboard visual 1: Executive Pulse. Screenshot of the authored Power BI report.

Evidence-led findings

Order outcomes. 250 Cancelled, 247 Returned, 237 Pending, 235 Shipped and 231 Delivered.

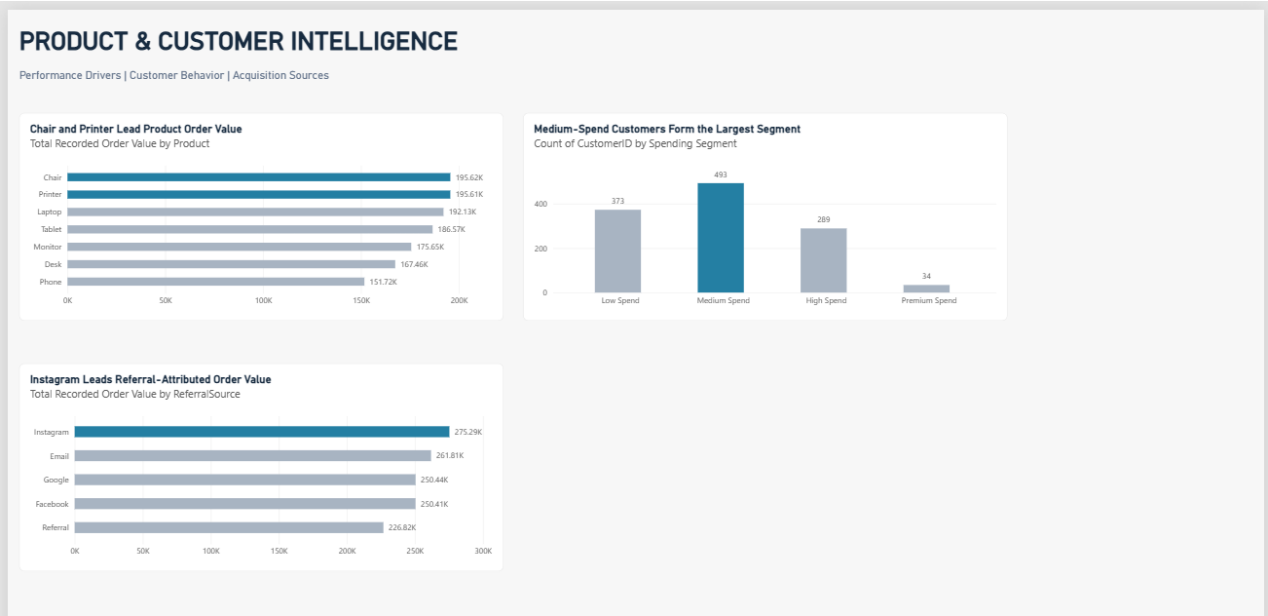
Peak month. June 2024 reaches 68,068.54 in recorded order value, the highest observed monthly total.

Interpretation. The trend communicates fluctuations in recorded value, while the status mix reveals a substantial share of affected orders.

03 / PRODUCT & CUSTOMER INTELLIGENCE

Who and what shape order value?

Which products, customers and sources deserve analytical attention?



Dashboard visual 2: Product & Customer Intelligence. Screenshot of the authored Power BI report.

Evidence-led findings

Product leaders. Chair (195,620.11) and Printer (195,612.61) are nearly tied in recorded order value.

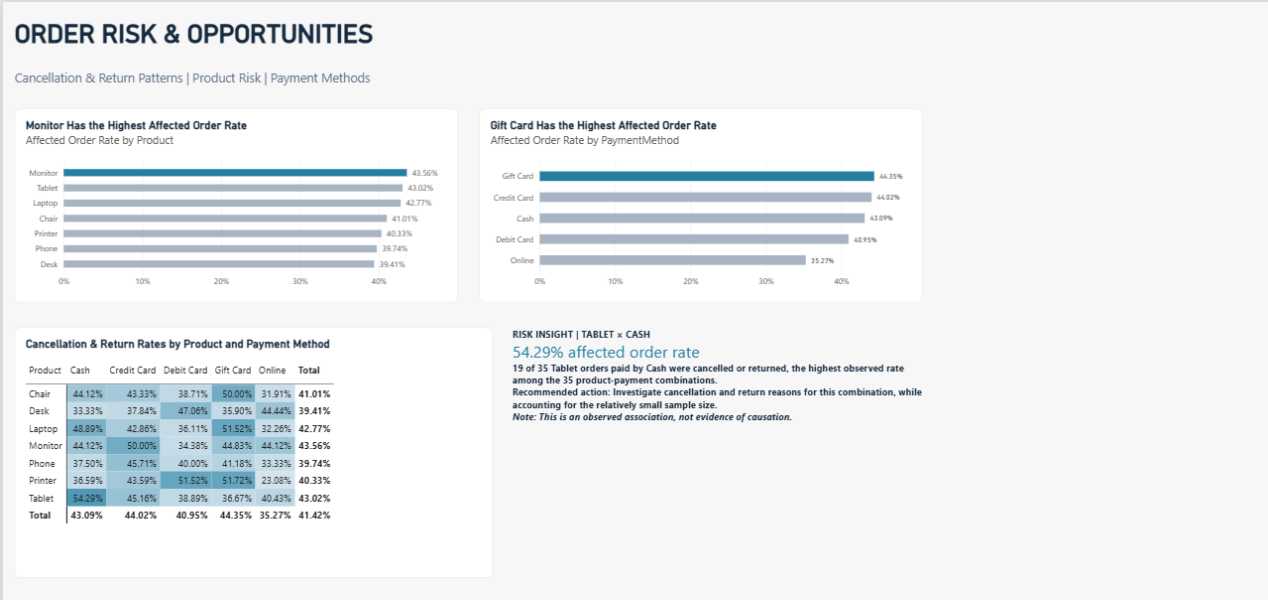
Customer composition. Low Spend: 373; Medium Spend: 493; High Spend: 289; Premium Spend: 34 customers.

Source signal. Instagram leads in recorded referral-attributed order value (275,285.45). This is not a measure of acquisition profitability.

04 / ORDER RISK & OPPORTUNITIES

Find the friction

Where are cancellations and returns comparatively concentrated?



Dashboard visual 3: Order Risk & Opportunities. Screenshot of the authored Power BI report.

Evidence-led findings

Product-level signal. Monitor has the highest observed affected order rate at 43.56%.

Payment-level signal. Gift Card has the highest observed affected order rate at 44.35%.

Intersection to investigate. Tablet × Cash has a 54.29% affected rate: 19 of 35 orders. A small subgroup and descriptive association.

05 / DESIGN & TECHNICAL METHOD

How the report was built

01 Frame the business problem Organize the dashboard around performance, commercial composition and operational risk instead of unrelated visuals.

02 Prepare transactional data Use the provided Excel order dataset in Power BI; examine dates, order status, product, payment method, customer identifier and recorded values.

03 Define reusable metrics Create DAX calculations for Total Orders, Total Recorded Order Value, Affected Orders and Affected Order Rate in filter context.

04 Build customer-level intelligence Construct a Customer Summary table with order count, recorded value and analyst-defined spending segments.

05 Model relationships Relate Customer Summary[CustomerID] to Orders[CustomerID] through an active one-to-many (1:*) relationship with single cross-filter direction.

06 Design the interface Use consistent navy and teal emphasis, compact KPI cards, descending comparisons, informative headings and a conditional-format risk matrix.

07 Check the outputs Reconcile headline totals and customer segment counts to prior analyses; verify report remains intact after relationship changes.

Definitions that make the report auditable

MEASURE	DEFINITION
Total Orders	Count of recorded order rows / IDs in current context.
Total Recorded Order Value	Sum of recorded TotalPrice across order statuses in current context.
Affected Orders	Count of records where OrderStatus is Cancelled or Returned.
Affected Order Rate	Affected Orders ÷ Total Orders, evaluated in current filter context.
Customer Spending Segment	Analyst-defined tier based on customer-level recorded order value.

06 / DECISION FRAMEWORK

What the business should do next

The purpose of this analysis is not to assign blame to a product or payment type. It is to establish a prioritized investigation plan supported by observed patterns.

PRIORITY	ACTION	EVIDENCE / PURPOSE
01	Diagnose cancellation and return reasons	41.42% of orders are affected. Collect reason codes and track stage of failure.
02	Investigate Monitor outcomes	Monitor has the highest product affected rate (43.56%). Compare cohort, delivery and quality data.
03	Review Gift Card customer journey	Gift Card has the highest payment affected rate (44.35%). Examine checkout, redemption and support cases.
04	Audit Tablet × Cash subgroup	19 of 35 orders affected (54.29%). Confirm sample stability and underlying causes before intervening.
05	Assess acquisition quality	Instagram leads recorded value, but source quality also requires outcome, cost and retention metrics.

A stronger next iteration

Measure outcomes. Add delivered net value, refunds, operational loss, actual payment settlement, and reason codes where available.

Add time context. Capture order creation, cancellation/return and delivery dates to distinguish process delay from cohort effects.

Validate signal robustness. Compare group sample sizes and trends over time, and test whether observed rate differences persist.

Close the loop. Link dashboard insights to an accountable operational review process with owners and follow-up actions.

07 / EVIDENCE & HANDOFF

Limitations, deliverables & reproducibility

Interpretation guardrails

- Recorded order value is not realized revenue, net sales or verified income. Cancelled, returned and pending orders are included in the recorded totals.
- An affected order is classified solely from the recorded OrderStatus. The dataset does not establish why or when that status occurred.
- Customer spending segments are analytical thresholds, not official business customer tiers.
- Referral source value describes the recorded attribution field and does not establish campaign ROI or incrementality.
- The highlighted Tablet × Cash combination has only 35 orders. Observed rates do not demonstrate statistical significance or causality.
- Some insight annotations on the report are static narratives describing the full dataset. They may not update when a user applies a filter.
- The relationship allows Customer Summary to filter Orders with single-direction filtering. The reverse interaction is not guaranteed.

Files prepared for handoff

ARTIFACT	PURPOSE
DecodeLabs_Project4_The_Order_Journey.pbix	Editable three-page Power BI report.
Dataset for Data Analytics (3).xlsx	Original transaction dataset.
ScreenShots/01_Executive_Pulse.png	Executive overview preview.
ScreenShots/02_Product_Customer_Intelligence.png	Customer and product preview.
ScreenShots/03_Order_Risk_Opportunities.png	Risk analysis preview.
README.md	Repository introduction, navigation and key results.
This case study (.docx)	Business-facing methods, findings and recommendations.

Reproduce and explore

Open the PBIX in a compatible Power BI Desktop version, navigate the three pages and clear selections to return to the overall view. If refresh fails after moving the PBIX, update the local Excel file path in Power Query to the included workbook.

Evidence basis: supplied Power BI dashboard screenshots and user-verified project results. No external revenue, marketing-cost, refund or causal evidence was available. Prepared for DecodeLabs Data Analytics Internship, Project 4.